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Music and Audio Solutions on Palm Powered™ Handhelds

Erika Mobley
Market Development Manager,
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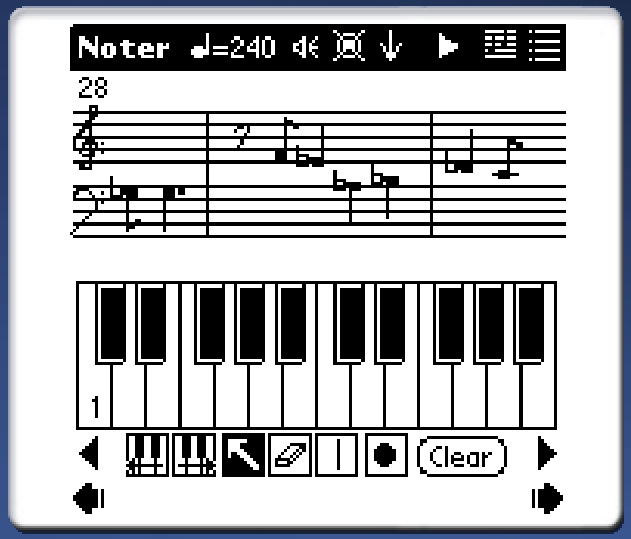
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Palm Music/Audio Overview: mp3

- “Better than… mp3”
- Music
- Non-music audio: audiobooks, news feeds
- Voice recording
- Mass storage for Enterprise
- Buzzwords: SDMI; DRM;DSP; AAC
- Prelude to future SD expansion devices

Palm Music/Audio Overview: MIDI





- Introducing Shinei's tsunamidi™:
  - Palm V™ MIDI hardware: your portable music studio
  - Composition, transcription and sequencing software from Minimusic™ and Tobel Studios™)
  - "Sync" a song

Here From Some Experts:

- Tom Harrah: Pocketpyro, Inc.
- K. C. Lim, Singapore Shinei Sangyo Corp.
- Chad Mealey, miniMusic, Inc.

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*More at today's Roundtable!! (#819 @ 4PM)

PocketPyro, Inc.

Tom Harrah
President

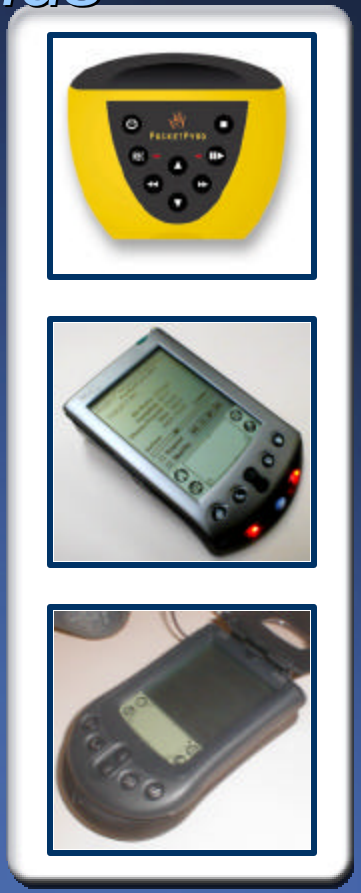


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DragonThunder

Hardware Products for Palm™ Handhelds

- III Series: “Pyro Sports Edition”
- V Series: “Pyro Professional Edition”
- m100: Shinei’s “Porteson”



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DragonThunder

Hardware Capabilities

- Powerful programmable DSP processing
- Internet Audio/Digital Music
- Audio record and playback
- Mass storage for existing Palm handhelds
- Memory expandable using SD/MMC cards
- Upgradeable to include
  - Synchronized Multi-Media
  - Text to Speech
  - Voice Recognition

Digital Audio Standards

- Music
  - MP3—Original and popular music compression
  - AAC—Advanced Audio Coding
  - WMA—Microsoft's compression standard
  - ATRAC3—Sony's Advanced MiniDisc format
- Voice
  - ADPCM—Simple voice compression
  - ACELP—Advanced voice compression

Digital Audio Standards

Typical Bit Rates

- Music (at or near CD quality)
  - CD – 1.4 Mbps
  - MP3 – 128 kbps
  - AAC – 96 kbps
  - WMA – 96 kbps
  - ATRAC3 – 128 kbps
- Voice (telephone quality)
  - Telephone – 64 kbps
  - ADPCM – 32 kbps
  - ACELP – 8 kbps

Digital Audio Standards

Typical File Sizes

- Music (1 hour at or near CD quality)
  - CD – 600 MBytes
  - MP3 – 60 MBytes
  - AAC – 45 MBytes
  - WMA – 45 MBytes
  - ATRAC3 – 60 MBytes
- Voice (1 hour at Telephone quality)
  - ADPCM – 15 MBytes
  - ACELP – 4 MBytes

Secure Digital Music Initiative (SDMI)

- Over 100 companies involved in developing the SDMI standards
- Goal is to enable a satisfying consumer experience with digital music while enforcing copyright protection
- Phase 1: Now Available: Prevents unauthorized copying or distribution
- Phase 2: Future Release: Will add screening to detect illegally distributed music

Digital Rights Management (DRM)

- DRM software implements security standards that may or may not be SDMI compatible
- All systems encrypt the content and implement rules for playing and copying the content
- Major vendors are:
  - Liquid Audio
    - Current Market Leader
    - Encrypts Music in any format into secure Liquid Audio format
  - Intertrust
    - Encrypts Music into ‘DigiBox’ format
  - IBM
    - Electronic Media Management System (EMMS)

Secure Digital (SD) Cards

- More advanced version of MMC Cards
- Cryptographic security based on proven security concepts from DVD audio
- Protection of copyrighted data
- A high data transfer rate for fast copy/download
- High storage capacity, 64MB and 128MB soon
- A user selectable mechanical write protect switch

DragonThunder Audio

More Than Just Music

- Audio books
  - Audible.com: www.audible.com
  - Mediabay.com: www.mediabay.com
- Audio News
  - Audiobasket: www.audiobasket.com
  - Voquette: www.voquette.com
- Audio Training
  - Audio Intelligence: www.audiointelligence.com
  - getAbstract: www.getabstract.com
- Games!!!

DragonThunder Software Development

- Free licensing of SDK for non-commercial applications
- Includes support for the new Palm SDK Virtual File System (VFS)
- Support for Play and Record of Multimedia content (audio initially)
- Flexible media I/O through support of “Media Contexts”
- Hardware independent for support of future devices
- Flexible and extensible for support of additional features and content types, such as synchronized multimedia

Hardware Availability

1st Quarter 2001 for All Palm Handheld Computers


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Audio Hardware Solutions

K.C. Lim
Business Development Director
Shinei International

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Agenda

- About Shinei International
- Objective
- Product Development Process
- List of products developed
- Other applications and opportunities
- Summary

About Shinei International

- Electro-mechanical contract manufacturer to Fortune top 50 IT companies
- Headquarters in Singapore with global presence in US, UK, China and Malaysia
- Electrical, Mechanical and Industrial Design capability in-house

Objective

- To share the presenter's experience in developing audio accessories for Palm™ handhelds
- To present opportunities for 3rd-party software or hardware enhancements

Producer Development Process

- Identify the opportunities
  - Overview of the Palm handheld's existing function set
  - Critique what could be done better
  - What else is missing that the user would like to have?
  - What is available on the other handhelds that's not on the Palm handhelds?

Product Development Process

- Product selection and definition
  - Market research of what's already out there
  - Pick the target market—e.g., consumer, education, enterprise, professional, etc.
  - Define the desired product feature set
  - Select the Palm handheld for the accessory

Product Development Process

- Cost and Time-to-Market analysis
  - Add packaging, distribution, warranty, marketing and tech support costs to the bill-of-materials
  - Factor in inventory holding and obsolescence costs
  - Factor in the EOL (end of life) of the handheld
  - Ballpark product development leadtime 6–9 months

Product Development Process

- Key success criteria
  - Form over function—don't underestimate importance of good industrial design—cool looks do sell
  - Strategic pricing—SRP less than the Palm™ handheld's
  - Simple, intuitive user interface

Shinei's Palm OS® Accessories

- Voice recorders
  - i-Vox for Palm III and V Series Handhelds
  - my-Vox for Visor/Prism
- Mobile entertainment devices
  - Tsunamidi MIDI synthesizer for Palm V
  - Porteson MP3 player for m100
  - my-Raydio FM module for Visor/Prism
- Others
  - Syncharger—cordless Palm V travel charger
  - clipmodem—landline modem for m100

Open for 3rd Party Enhancements

Examples

- From voice recorder to phone dialer to assertive speech device
- From MIDI player to music content downloads and game sounds on the Palm handheld
- From MP3 player to high capacity data storage, e-books, speech-to-text conversion
- HDKs out of syncharger, modem casings

Summary

- Functions not native to handheld = developers' opportunities
- Audio is the new frontier of mobile entertainment and productivity enhancement on the Palm handhelds
- Open your add-ons to 3rd party aftermarket developments = win-win


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Adding MIDI to Palm OS® Solutions

Chad Mealey
President, miniMusic

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Presentation Overview

- What is MIDI?
- Advantages of using MIDI on Palm OS
- miniMusic applications
- MIDI hardware for the Palm Platform
- Adding MIDI to your Palm OS application
- Contacting miniMusic

What Is MIDI
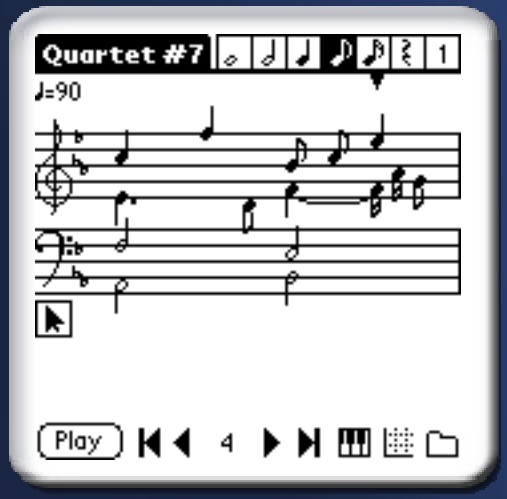
- MIDI = Musical Instrument Digital Interface
- Protocol created in 1982
- Not digital sound like MP3 or WAV
- Control electronic music instruments (and lights, sound equipment, etc.)
- “General MIDI” assures compatibility

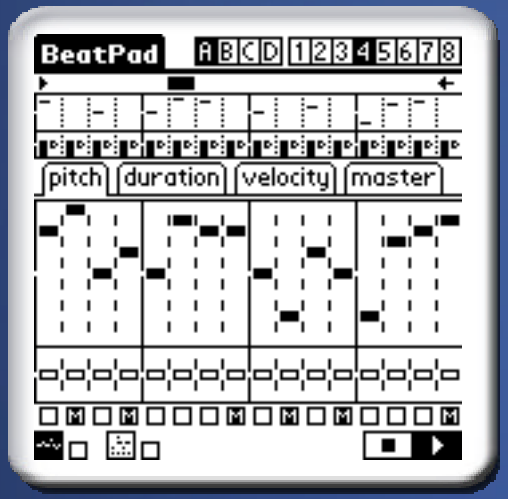
Advantages of MIDI on Palm OS®

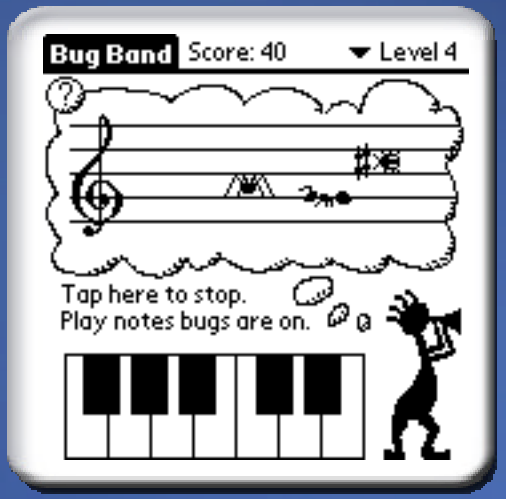
- MIDI already used in the Palm OS (since 3.0)
- Industry standard
- Easy to set up with serial port
- Low processor requirements
- Allows interactivity (unlike MP3)
- Expands the sound capabilities now!

miniMusic Applications

- NotePad: Notation sequencer
  - Create, edit and play music
- BeatPad: Pattern sequencer
  - Edit patterns as they play!
- BugBand: Musical edu-tainment







Demonstration

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MIDI Hardware for the Palm Platform

- Tsunamidi by Shinei
  - Palm V style sled
  - Built-in speaker and headphone jack
  - MIDI in/out cable
  - www.tsunamidi.com
- SG20 by Swivel Systems
  - Palm III style “modem” case
  - Headphone Jack only
  - Volume control and bass boost
  - MIDI in/out + RCA cable
  - www.swivelsystems.com

Adding MIDI to Palm OS® Software

- Palm OS sound manager routines only work with speaker
- Set serial speed to 31250 baud
- Speaker and serial port drain battery when on
- MIDI hardware may need time to warm up
- Give user choice of speaker or MIDI

Opening Serial Port

```
static void StartApplication(void)
{
    Err             err1;
    UInt            port = 0;
    ULong           baud = 31250;

   // Open port if preference is set.
   if (Prefs.flags.midi){
      err1 = SysLibFind("Serial Library",
         &Refno);
      if (err1 == 0){
         err1 = SerOpen(Refno,port,baud);
      }
   }
}
```


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Closing Serial Port

```
static void StopApplication(void)
{
   Err     error;

   // Close port if preference is set.
   if (Prefs.flags.midi){
      // Turn off serial port.
      error = SerSendWait(Refno,-1);
      error = SerClose(Refno);
   }
   // SAVE APPLICATION PREFERENCES
   // Close all open forms
}
```

MIDI Commands

- Format:
  - c = Channel (0-15)
  - p = Program (0-127)
  - v = Velocity (0-127)
  - n = Note Value (0-127), 60 is “middle c”
- Note On: **1001cccc 0nnnnnnn 0vvvvvvv**
- Note Off: **1000cccc 0nnnnnnn 0vvvvvvv**
- Program Change: **1100cccc 0ppppppp**

- Also Pitch Bend, Bank Change, Volume, etc.

Contacting miniMusic

Chad Mealey, President

415-831-1514

chad@minimusic.com

Developer page:

www.minimusic.com/dev

All we ask: Add MIDI to your software!!!


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